Course: BCSNT 6063 – Object Oriented Programming (2nd Semester)

Assignment Type: Practical / Tutorial

Submission: Screenshots + Program Files

1.Smart attendance checker

SmartAttendanceChecker.java

```java
import java.util.Scanner;

public class SmartAttendanceChecker {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter total classes conducted: ");
        int totalClasses = sc.nextInt();

        System.out.print("Enter classes attended: ");
        int attended = sc.nextInt();

        double percentage = ((double) attended / totalClasses) * 100;

        System.out.println("Attendance Percentage: " + percentage + "%");

        if (percentage >= 75) {
            System.out.println("Eligible for Exam");
        } else {
            System.out.println("Not Eligible");
        }

        sc.close();
    }
}
```

Run SmartAttendanceChecker

```
C:\Users\LOQ\.jdks\openjdk-26.0.1\bin\java.exe "-javaa
Enter total classes conducted: 3
Enter classes attended: 3
Attendance Percentage: 100.0%
Eligible for Exam
#
Process finished with exit code 0
```

2. AI Age Predictor

SmartAttendanceChecker.java AIAgePredictor.java

```java
import java.time.Year;
import java.util.Scanner;

public class AIAgePredictor {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter your current age: ");
        int age = sc.nextInt();

        System.out.println("Age after 10 years: " + (age + 10));
        System.out.println("Age after 25 years: " + (age + 25));
        System.out.println("Age after 50 years: " + (age + 50));

        int currentYear = Year.now().getValue();
        int yearTurning100 = currentYear + (100 - age);

        System.out.println("You will turn 100 in: " + yearTurning100);

        sc.close();
    }
}
```

Run AiAgePredictor

```
C:\Users\LOQ\.jdks\openjdk-26.0.1\bin\java.exe "-javaagent:
Enter your current age: 23
Age after 10 years: 33
Age after 25 years: 48
Age after 50 years: 73
You will turn 100 in: 2103

Process finished with exit code 0
```

3. Fitness Challenge Tracker

SmartAttendanceChecker.java AiAgePredictor.java FitnessChallenge.java

```java
import java.util.Scanner;

public class FitnessChallenge {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int total = 0;
        int highest = 0;

        for (int i = 1; i <= 7; i++) {
            System.out.print("Enter steps for day " + i + ": ");
            int steps = sc.nextInt();

            total += steps;

            if (steps > highest) {
                highest = steps;
            }
        }

        double average = total / 7.0;

        System.out.println("Total Steps: " + total);
        System.out.println("Average Steps: " + average);
        System.out.println("Highest Steps: " + highest);

        sc.close();
    }
}
```

```
Run   FitnessChallenge ×

C:\Users\LOQ\.jdks\openjdk-26.0.1\bin\java.exe "-javaagent:C
Enter steps for day 1: 43
Enter steps for day 2: 4344
Enter steps for day 3: 343
Enter steps for day 4: 434
Enter steps for day 5: 343
Enter steps for day 6: 4343
Enter steps for day 7: 443
Total Steps: 10293
Average Steps: 1470.4285714285713
Highest Steps: 4344
```

4. Fibonacci Series

SmartAttendanceChecker.java AiAgePredictor.java FitnessChallenge.j

```java
import java.util.Scanner;

public class Fibonacciseries {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of terms: ");
        int n = sc.nextInt();

        int first = 0, second = 1;

        System.out.println("Fibonacci Series:");

        for (int i = 1; i <= n; i++) {
            System.out.print(first + " ");

            int next = first + second;
            first = second;
            second = next;
        }

        sc.close();
    }
}
```

Run Fibonacciseries

```
C:\Users\LOQ\.jdks\openjdk-26.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.3\lib\idea_rt.jar=
Enter number of terms: 1
Fibonacci Series:
0
Process finished with exit code 0
```

Performance

Show Results

5) Prime Number Check

```java
import java.util.Scanner;

public class PrimeNumberCheck {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int number = sc.nextInt();

        boolean isPrime = true;

        if (number <= 1) {
            isPrime = false;
        } else {
            for (int i = 2; i <= number / 2; i++) {
                if (number % i == 0) {
                    isPrime = false;
                    break;
                }
            }
        }

        if (isPrime) {
            System.out.println(number + " is Prime");
        } else {
            System.out.println(number + " is Not Prime");
        }

        sc.close();
```

Run PrimeNumberCheck

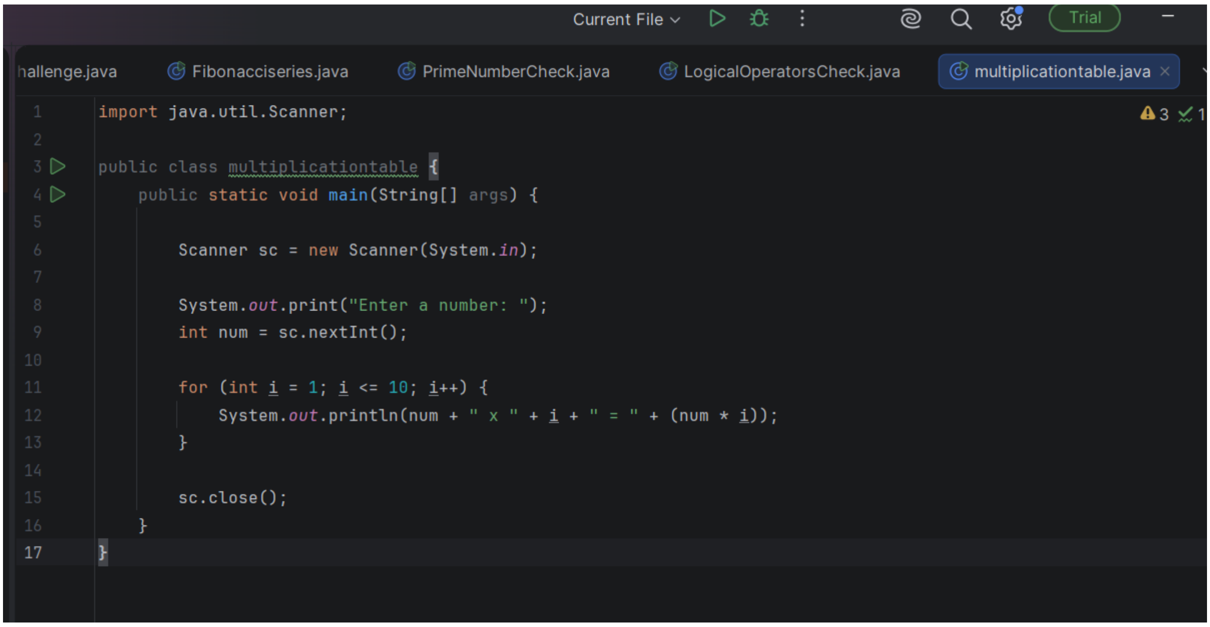
```
C:\Users\LOQ\.jdks\openjdk-26.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.3\lib\idea_rt.jar=
Enter a number: 1
1 is Not Prime

Process finished with exit code 0
```

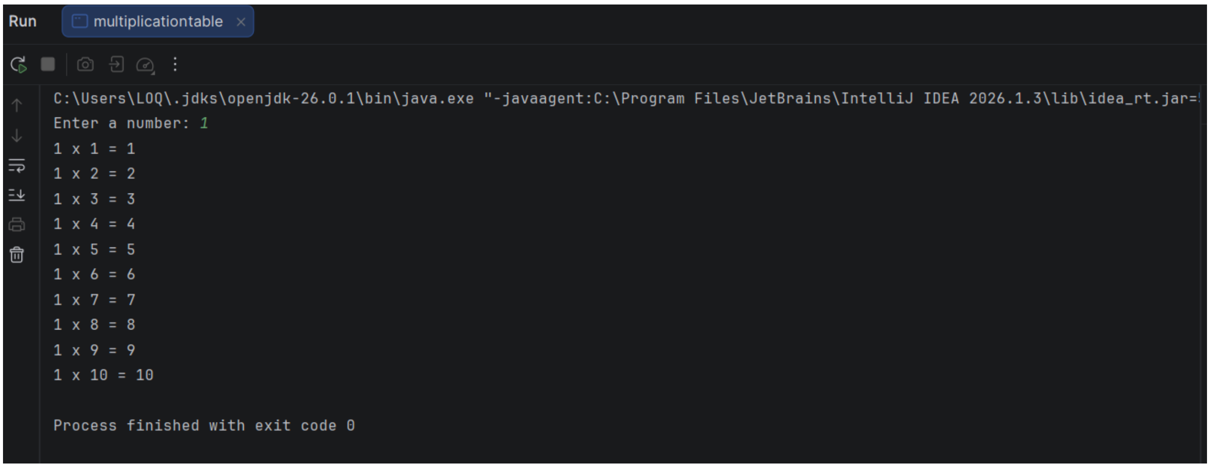
6) Logical Operators Check

```
import java.util.Scanner;

public class LogicalOperatorsCheck {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int a = sc.nextInt();

        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        System.out.print("Enter third number: ");
        int c = sc.nextInt();

        boolean result = (a + b == c);

        System.out.println("Is third number equal to sum of first two? " + result);

        sc.close();
    }
}
```

```
C:\Users\LOQ\.jdks\openjdk-26.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.3\lib\idea_rt.jar=
Enter first number: 1
Enter second number: 2
Enter third number: 3
Is third number equal to sum of first two? true

Process finished with exit code 0
```

7) Multiplication Table





8) Right-Angled Triangle Pattern

PrimeNumberCheck.java LogicalOperatorsCheck.java multiplicationta

```java
import java.util.Scanner;

public class RightAngledTrianglePattern {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of rows: ");
        int rows = sc.nextInt();

        for (int i = 1; i <= rows; i++) {

            for (int j = 1; j <= i; j++) {
                System.out.print("*");
            }

            System.out.println();
        }

        sc.close();
    }
}
```

```
C:\Users\LOQ\.jdks\openjdk-26.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.3\lib\idea_rt.jar=
Enter number of rows: 5
*
**
***
****
*****

Process finished with exit code 0
```

9) Pyramid Pattern

LogicalOperatorsCheck.java multiplicationtable.java RightAngledTrianglePattern.java C:\Users\LOQ\Id

```java
import java.util.Scanner;

public class PyramidPattern {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of rows: ");
        int rows = sc.nextInt();

        for (int i = 1; i <= rows; i++) {

            for (int j = 1; j <= rows - i; j++) {
                System.out.print(" ");
            }

            for (int k = 1; k <= (2 * i - 1); k++) {
                System.out.print("*");
            }

            System.out.println();
        }

        sc.close();
    }
}
```

Run PyramidPattern

```
C:\Users\LOQ\.jdks\openjdk-26.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.3\lib\idea_rt.jar=
Enter number of rows: 6
     *
    ***
   *****
  *******
 *********
***********

Process finished with exit code 0
```

10) Palindrome Number

```
multiplicationtable.java    RightAngledTrianglePattern.java    PyramidPattern.jav

import java.util.Scanner;

public class PalindromeNumber {

    public static boolean isPalindrome(int x) {  1 usage

        int original = x;
        int reversed = 0;

        while (x != 0) {
            int digit = x % 10;
            reversed = reversed * 10 + digit;
            x = x / 10;
        }

        return original == reversed;
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int number = sc.nextInt();

        System.out.println(isPalindrome(number));

        sc.close();
    }
}
```

```
C:\Users\LOQ\.jdks\openjdk-26.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.3\lib\idea_rt.jar=
Enter a number: 6
true

Process finished with exit code 0
```

11) Basic Calculator

```java
import java.util.Scanner;

public class BasicCalculator {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        double a = sc.nextDouble();

        System.out.print("Enter second number: ");
        double b = sc.nextDouble();

        System.out.print("Enter operator (+, -, *, /): ");
        char operator = sc.next().charAt(0);

        double result = 0;

        switch (operator) {

            case '+':
                result = a + b;
                break;

            case '-':
                result = a - b;
                break;

            case '*':
```

```java
                break;

            case '*':
                result = a * b;
                break;

            case '/':
                if (b != 0) {
                    result = a / b;
                } else {
                    System.out.println("Cannot divide by zero");
                    return;
                }
                break;

            default:
                System.out.println("Invalid operator");
                return;
        }

        System.out.println("Result: " + result);

        sc.close();
    }
}
```

```
C:\Users\LOQ\.jdks\openjdk-26.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.3\lib\idea_rt.jar=
Enter first number: 9
Enter second number: 99
Enter operator (+, -, *, /): +
Result: 108.0

Process finished with exit code 0
```

12) Grade Calculator

```java
import java.util.Scanner;

public class Gradecalculator {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter marks: ");
        int marks = sc.nextInt();

        String grade =
                (marks >= 90) ? "A+" :
                        (marks >= 80) ? "A" :
                        (marks >= 70) ? "B+" :
                        (marks >= 60) ? "B" :
                        (marks >= 50) ? "C+" :
                        (marks >= 40) ? "C" :
                        "F";

        System.out.println("Grade: " + grade);

        sc.close();
    }
}
```

```
C:\Users\LOQ\.jdks\openjdk-26.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.3\lib\idea_rt.jar=
Enter marks: 4
Grade: F

Process finished with exit code 0
```